

EMMA SOFIA SOSA

email: sosa45nh@gmail.com

ORCID: 0000-0002-1765-6191

website: emmasosa.com

EDUCATION

- California Institute of Technology**, Pasadena, CA 2019 – 2024
PhD in Geology
Thesis: [Composition, Structure, and Formation of the Lower Crust in Continental and Oceanic Arc Settings: Insights from the Xenolith Record](#)
- California Institute of Technology**, Pasadena, CA 2017 – 2019
Master's in Geology
- Lafayette College**, Easton, PA 2013 – 2017
Bachelor of Science in Geology, Magna Cum Laude

PROFESSIONAL EXPERIENCES

- Analytical Scientist, [CREW Carbon](#), New Haven, CT 2025 – present
Graduate Researcher, California Institute of Technology, Pasadena, CA 2017 – 2024
Undergraduate Research Fellow, Smithsonian Institution, NMNH, Washington, D.C. Summer 2016
EXCEL Research Scholar: Lafayette College, Easton, PA Summer 2015

SELECTED AWARDS

- AAAS Science & Technology Policy Fellowships: Executive Branch**, Finalist (*declined*) Winter 2025
Selected from a national applicant pool of STEM PhDs at all career stages for consideration as a scientific advisor within a U.S. federal agency; declined prior to placement interviews due to uncertainty in agency funding.
- USAID Science for Development Fellowships**, Semi-Finalist (*program terminated*) Winter 2025
One of 20 semi-finalists selected from >100 applicants for a program embedding PhD scientists in international development programming; program terminated before interviews.
- Gunilla Hastrup Adventure Award**, The Caltech Y Fall 2023
Supported five-week intensive Spanish language program in Costa Rica (4+ hours daily instruction, homestay immersion).
- Graduate Research Fellowship Program (GRFP)**, National Science Foundation Spring 2019
Three-year merit-based award covering Caltech graduate student tuition and research stipend.
- Earle C. Anthony Fellowship**, Caltech Spring 2017
One-year merit-based fellowship awarded to an incoming STEM graduate student from an underrepresented background.
- Dwornik Award**, Geological Society of America Spring 2017
Best undergraduate poster presentation, Lunar and Planetary Science Conference (LPSC), 2017.
- Guy and Joyce Hovis Geology Award**, Lafayette College Spring 2016
Recognition of achievement in undergraduate research.
- Stephen G. Pollock Undergraduate Research Grant**, Geological Society of America Spring 2015
Funded fieldwork in Alaska collecting tephra samples for undergraduate research project; results presented at GSA 2016.

PUBLICATIONS (PEER-REVIEWED)

- Sosa, E. S., Bucholz, C. E., Hernández-Montenegro, J. D., Rodríguez-Vargas, A., Kipp, M. A., & Tissot, F. L. H. (2024b). Garnet clinopyroxenite formation via amphibole-dehydration in continental arcs: Evidence from Fe isotopes. *Earth and Planetary Science Letters*, 648, 119050.
- Sosa, E. S., Bucholz, C. E., Hernández-Montenegro, J. D., Kipp, M. A., Tissot, F. L. H., Ratschbacher, B. C., Jackson, J. M., Kay, S. M., & Kay, R. W. (2024a). Lower crustal control in the iron isotope systematics of plutonic xenoliths from Adak Island, Central Aleutians, with implications for arc magma geochemistry. *Geochimica et Cosmochimica Acta*, 377, 1-18.

Hernández-Montenegro, J. D., Bucholz, C. E., **Sosa, E. S.**, Kipp, M. A., & Tissot, F. L. H. (2024). Iron isotope fractionation during partial melting of metapelites and the generation of strongly peraluminous granites: a case study of the Neoarchean Ghost Lake batholith. *Geochimica et Cosmochimica Acta*, 380, 112-130.

Sosa, E. S., Bucholz, C. E., Barickman, M. H., VanTongeren, J. A., Setera, J. B., Kay, S. M., & Kay, R. W. (2023). Petrology and geochemistry of Adak Island plutonic xenoliths: implications for primitive magma generation and crustal differentiation in the Aleutian Island arc. *Journal of Petrology*, 64(10), egad073.

Lunning, N. G., Gardner-Vandy, K. G., **Sosa, E. S.**, McCoy, T. J., Bullock, E. S., & Corrigan, C. M. (2017). Partial melting of oxidized planetesimals: An experimental study to test the formation of oligoclase-rich achondrites Graves Nunataks 06128 and 06129. *Geochimica et Cosmochimica Acta*, 214, 73-85.

SELECTED RESEARCH EXPERIENCES

Wastewater process improvement and decarbonization with CaCO₃ dosing: CREW Carbon 2025 – present
Conducting collaborative research on how calcite (CaCO₃) dosing improves wastewater treatment through better sludge dewatering, reduced chemical use, and increased nutrient removal while capturing CO₂. Preliminary results presented at the [2026 Residuals, Biosolids, and Treatment Technology Conference](#) in Kansas City, MO. Leading the physicochemical characterization of CaCO₃ feedstocks to predict which products show the most advantageous properties for efficient mineral dissolution and communicating results to the leadership team. Carbon-removal measurement, reporting, and verification (MRV) and data analysis lead for five sites.

Fe isotope fractionation during densification of metamorphic xenoliths: Caltech 2022 – 2024
Separated and purified iron (Fe) from bulk-rock powders and mineral separates (garnet, amphibole, pyroxene) from mid/lower-crustal Andean xenoliths ("arclogites"). Combined thermodynamic Perple_X modeling with fundamental isotope theory to show that many garnet clinopyroxenites in the lower Andean crust likely formed through amphibole dehydration reactions. Integrated Fe isotope results with major/trace element mineral chemistry, geothermobarometry, petrography, and whole-rock chemistry to develop and support conclusions. Results published in [Earth and Planetary Science Letters](#).

Fe isotope systematics of lower-crustal igneous xenoliths: Caltech 2020 – 2024
Separated and purified Fe from bulk-rock powders and mineral separates (olivine, pyroxene, spinel, magnetite, amphibole) from mid/lower-crustal Aleutian Arc igneous xenoliths. Used mass-balance modeling to show how magmas may experience periods of isotopic enrichment or depletion as the crystallizing mineral assemblage changes. Established the laboratory and analytical procedures for Fe isotope analysis at Caltech, including data reduction codes, reference material procurement and preparation, and cross-calibration of standards against external labs. Findings were presented at the 2022 Goldschmidt Conference in Honolulu, HI. Results published in [Geochimica et Cosmochimica Acta](#).

Crystallization conditions at the base of the Aleutian Arc crust: Caltech 2018 – 2023
Collected in situ mineral trace and major element data to estimate the crystallization conditions (pressure, temperature, water content, oxygen fugacity) under which Aleutian Arc xenoliths formed. Created a set of secondary spinel EPMA standards with Fe³⁺/ΣFe ratios independently measured by synchrotron Mössbauer spectroscopy, enabling more accurate Fe speciation assessment of Fe-Ti oxides for future oxybarometry studies at Caltech. Results published in [Journal of Petrology](#).

Fe³⁺/ΣFe ratios in amphibole: Caltech Summer 2018
Assisted a Caltech postdoc in performing single-crystal synchrotron Mössbauer spectroscopy at Argonne National Laboratory. Data were used to calculate Fe³⁺/ΣFe ratios in amphibole grains and later employed in Sosa et al. (2024a, GCA) to estimate the force constant of Fe bonds in the amphibole lattice structure.

Senior Honors Thesis: Lafayette College 2016 – 2017
Used zircon grains separated from outwash deposits to study the glacially obscured Öraefajökull volcano in southeastern Iceland. Collected trace element data from zircon at Stanford University using sensitive high-resolution ion microprobe-reverse geometry (SHRIMP-RG). Results were presented at the 2017 GSA Joint Section Meeting in Pittsburgh, PA.

Natural History Research Experiences (NSF-REU) Fellow: Smithsonian Institution, NMNH Summer 2016
Performed partial melting experiments on chondrites to explore temperature and oxygen fugacity conditions under which the oligoclase-rich achondrites Graves Nunataks 06128 and 06129 formed. Findings presented at the Council for Undergraduate Research REU symposium at the National Science Foundation headquarters in Alexandria, Virginia, and the 2017 Lunar and Planetary Science Conference in Houston, TX. Results published in [Geochimica et Cosmochimica Acta](#).

EXCEL Research Scholar: Lafayette College

Summer 2015

Assisted in developing a project using zircon trace element and isotope geochemistry to link orphaned ash beds to their source volcanoes. Established laboratory protocols for zircon mineral separation at Lafayette College. Results were presented at the GSA Annual Meeting in Denver, CO.

METHODOLOGICAL EXPERTISE

Stable isotope geochemistry and mass spectrometry: ultra-clean laboratory techniques, including Fe isotope separation and purification via ion-exchange chromatography, acid digestions of mineral separates (including Parr bomb), high-precision isotope ratio measurement via MC-ICP-MS (Thermo Fisher Neptune Plus).

Elemental analysis: bulk and in situ measurements via ICP-OES, EPMA, SEM(-EDS), SIMS, whole-rock XRF, micro-XRF mapping, solution and laser-ablation ICP-MS.

Mineralogical characterization: powder XRD, BET surface area analysis (Anton Paar Nova 800), laser diffraction particle size analysis, optical microscopy, mineral separations (zircon, garnet, Fe-Ti oxides, pyroxene, olivine, amphibole) through magnetic and density techniques and hand-picking.

Volatile element analysis: high-temperature combustion total carbon (TC) and sulfur analysis, total inorganic carbon (TIC) analysis through acidification (Eltra CS Analyzer).

SELECTED TEACHING AND MENTORING

Teaching assistant and laboratory instructor: duties included designing lab exercises for students, leading laboratory instruction, grading reports and homework, chaperoning field trips, and creating lab finals.

Metamorphic Petrology (Caltech)

Spring 2022

Structural Geology (Lafayette College)

Fall 2016

Introduction to Geology (Lafayette College)

Fall 2014

Research mentor/laboratory supervisor: taught students and interns laboratory techniques, helped them define the scopes of their projects/responsibilities, derive data-driven process insights, write reports, and create research presentations.

Renee Ma: intern (CREW Carbon)

Winter 2026 – present

Odatha (Odie) Kotagama: intern/Carbon Removal Associate Scientist (CREW Carbon)

Summer 2025 – present

Odalys Callejas: WAVE undergraduate research fellow (Caltech)

Summer 2021

Allyson Trussell: undergraduate research assistant (Caltech)

Fall 2017

SELECTED SERVICE, LEADERSHIP, AND OUTREACH EXPERIENCES

Science team representative: CREW Social Committee, CREW Carbon

2026 – present

Science team representative for an internal company social committee, organizing events to foster team bonding and culture.

Geology lead: lunch-and-learn seminar series, CREW Carbon

2026 – present

Leading an internal geology education program to build cross-functional technical fluency in the geosciences.

Event volunteer, Armory Center for the Arts, Pasadena

2024 – 2025

Assistant art teacher for community outreach events.

Union Station Adopt-a-meal, Caltech Y

2023 – 2025

Monthly meal service for 30–40 residents of Union Station Homeless Services; served as both volunteer and event lead.

Invited speaker, Geological and Planetary Sciences (GPS) Open House, Caltech

Spring 2023

Inaugural speaker for divisional community engagement program; spoke to children ages 6+ about art and science at Caltech.

Caltech Letters Illustrator, Caltech

2020 – 2021

Illustrator for the science communication publication [Caltech Letters](#).

Chair, Caltech Arts Committee, Caltech Graduate Student Council

2019 – 2022

Organized yearly exhibition of Caltech community art (undergrad, grad students, postdocs, faculty, staff), including event advertising, assembling a submission review committee, arranging collection/return of community art, sizing and printing digital art, installing the gallery, managing event volunteers, and organizing an opening night gala for the exhibit. Designed a website to host a virtual art gallery during the COVID-19 pandemic: [Caltech Community Artists](#).

- March for Science**, Caltech *Spring 2018*
Ran booth for day-long educational outreach event at Caltech. Supervised members of the public holding meteorites and answered questions about how meteorites are used to study igneous processes in the solar system.
- Project Scientist activity leader, meteoritics**, Caltech *Spring 2018*
Led meteorite lectures and activities for elementary and middle school girls at a departmental women-in-STEM outreach day.
- Member of Caltech Women in GPS (WINGS)**, Caltech *2017 – 2023*
Departmental advocacy group for women in geosciences; led a coffee-hour series bringing guest speakers and students together to discuss women's issues in academia.
- Natural History Research Experiences (NSF-REU) outreach day**, Smithsonian *Summer 2016*
Day-long educational outreach event at the Smithsonian National Museum of Natural History, allowing museum visitors to hold and interact with meteorites while educating them about the research done at the Smithsonian.
- Geology Club Secretary**, Lafayette College *2015 – 2016*
Coordinated activities and programs related to the geological sciences.

SELECTED FIELD EXPERIENCES

- Metamorphic geology field mapping**, Assynt, Scotland (2 weeks, University of Oxford) *Spring 2022*
- Sedimentary and structural geology field mapping**, Wheeler Ridge, AZ (10 days, Caltech) *Spring 2019*
- Structural geology field mapping**, Bishop, CA (10 days, Caltech) *Fall 2018*
- Amargosa River hydrologic field survey**, Death Valley, CA (10 days, Caltech) *Fall 2017*
- Structural geology field mapping**, Sheep Mountain, WY (5 days, Lafayette College) *Fall 2015*
- Volcanic ash collection**, Fairbanks, AK (5 days, Lafayette College) *Summer 2015*
- Field class, the Andes and Galápagos Islands**, Ecuador (3 weeks, Lafayette College) *Winter 2015*

CONFERENCE ABSTRACTS

- Residuals, Biosolids, and Treatment Technology Conference 2026: Kossakowski, J., Geyman, B., Vu, K., Katchinoff, J., Popolizio, N., **Sosa, E. S.**, Kotagama, O., Planavsky, N. (2026) CaCO₃ densifies activated sludge to improve wet- and cold-season solids separation at scale.
- AGU Fall Meeting 2022: Ratschbacher, B. C., **Sosa, E. S.**, Bucholz, C. E., Toellner, T. (2022) Characterization of lower crustal arc cumulate amphiboles (H₂O content, δD , Fe³⁺/Fe^T ratios) by a combination of microanalytical techniques on single crystals.
- Goldschmidt Conference 2022: **Sosa, E. S.**, Bucholz, C. E., Hernández-Montenegro, J. D., Tissot, F. L. H., Kipp, M. A., Kay, S. M., Kay, R. W. (2022) Iron isotopes in mantle and cumulate xenoliths from Adak Island, Central Aleutians.
- Goldschmidt Conference 2022: Bucholz, C. E., Hernández-Montenegro, J. D., **Sosa, E. S.**, Tissot, F. L. H., Kipp, M. A. (2022) Fractionation of iron isotopes between strongly peraluminous granites and their sedimentary sources: a case study of the Archean Ghost Lake batholith.
- Cordilleran GSA 2017: Carley, T. L., **Sosa, E. S.**, Banik, T. J., Coble, M. A., Fisher, C. M., Miller, C. F. (2017) Combined insight from tephra and glacially-derived sediments: A zircon-based study of Öräfajökull, SE Iceland.
- IAVCEI Volcano Mapping Workshop 2017: Carley, T. L., Banik, T. J., **Sosa, E. S.**, Coble, M. A. (2017) Altered, eroded, obscured by ice: A detrital zircon perspective into Iceland's rhyolitic history.
- NE GSA 2017: DeBarba, W. P., Carley, T. L., Pamukcu, A. S., Banik, T. J., **Sosa, E. S.**, Tintle, L. R. (2017) Geothermobarometry of glacial outwash sediment from Öräfajökull, Iceland.
- NE GSA 2017: **Sosa, E. S.**, Carley, T. L., Banik, T. J., Claiborne, L. L., Miller, C. F. (2017) A detrital zircon investigation into the magmatic history of the Öräfajökull volcano in Southeastern Iceland.
- LPSC 2017: **Sosa, E. S.**, Lunning, N. G., McCoy, T. J., Bullock, E. S., Corrigan, C. M., Gardner-Vandy, K. G. (2017) Constraining the petrogenesis of the paired achondrites GRA 06128/9 through partial melting of an oxidized chondrite.
- GSA 2016: **Sosa, E. S.**, Carley, T. L., Jensen, B. J. L., Coble, M. A. (2016) A zircon-based investigation into the tephra of uncertain origins: A case study with the Old Crow Tephra in Central Alaska.